```
while(!pred())
{
    wait(lock);
}
```

Throws

Any exception thrown by a call to `pred`, or `std::system_error` if the effects could not be achieved.

Note

The potential for spurious wakeups means that it's unspecified how many times `pred` will be called. `pred` will always be invoked with the mutex referenced by `lock` locked, and the function shall return if (and only if) an evaluation of `(bool)pred()` returns `true`.

Synchronization

Calls to `notify_one(), notify_all(), wait(), wait_for()`, and `wait_until()` on a single `std::condition_variable_any` instance are serialized. A call to `notify_one()` or `notify_all()` will only wake threads that started waiting *prior* to that call.

Std::Condition_Variable_Any::Wait_For Member Function

Waits until the `std::condition_variable_any` is notified by a call to `notify_one()` or `notify_all()` or until a specified time period has elapsed or the thread is woken spuriously.

Declaration

```
template<typename Lockable,typename Rep,typename Period>
std::cv_status wait_for(
```